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Adaptive Cross-Modal UAV Semantic Communications via Hierarchical DRL

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Abstract

This paper investigates cross-modal task-oriented semantic communication for a low-altitude unmanned aerial vehicle (UAV) transmitting mission-relevant semantic packets to a ground station over a wireless uplink. Different from bit-level transmission, the UAV first extracts task-relevant semantics from heterogeneous onboard sensors and then adaptively determines the sensing modality combination, semantic granularity, and transmit power according to channel variation and mission urgency. To capture the long-term coupling among semantic utility, energy consumption, and delay, the problem is formulated as a Markov decision process (MDP) that maximizes the long-term average semantic utility subject to average energy and delay constraints. A hierarchical deep reinforcement learning (DRL) framework is developed, where a high-level recurrent proximal policy optimization (PPO) policy selects the cross-modal semantic mode and semantic granularity, and a low-level soft actor-critic (SAC) policy determines the transmit power. In addition, a task-oriented semantic encoder is designed with cross-modal alignment loss, multi-granularity consistency loss, task loss, quantization loss, and rate regularization. The proposed framework enables semantic-level decision making rather than merely physical-layer resource control. Extended simulation results show that the proposed hierarchical DRLbased scheme outperforms several baselines in terms of semantic utility, energy consumption and delay.

Keywords: UAV communications, semantic communications, task-oriented communications, cross-modal sensing, hierarchical deep reinforcement learning.

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1 Introduction

Low-altitude unmanned aerial vehicle (UAV) networks can support emergency inspection, search and rescue [1], temporary surveillance [2], and mission-oriented environment understanding [3]. In these scenarios, an UAV often carries heterogeneous sensors such as RGB cameras, thermal cameras, or lightweight range sensors. However, transmitting all sensed raw data is usually unnecessary and inefficient, especially when the wireless channel is time-varying and the UAV operates under stringent battery and delay budgets [4]. What matters for the receiver is not exact bit-level fidelity, but whether the received information is sufficient to complete the target task [5].

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This observation naturally motivates task-oriented semantic communication, in which the transmitter extracts and conveys compact semantics that are directly relevant to the mission [6, 7]. Nevertheless, applying semantic communication to lowaltitude UAV systems faces three challenges [8]. First, the usefulness of each sensing modality changes dynamically with the environment. For instance, RGB is informative in clear daytime conditions, while thermal sensing is more reliable at night [2]. Second, semantic packet size is also task dependent: coarse semantics can be sufficient under urgent or poor-channel situations, while fine semantics may only be worthwhile when channel and battery conditions permit [9]. Third, energy and delay constraints are long-term and coupled, which makes myopic slot-by-slot optimization inadequate [10, 11].

The resulting online planning problem is difficult to handle by conventional optimization methods. In particular, the semantic mode and semantic granularity must be selected based on mission context and modality quality, while the transmit power should be adapted to the instantaneous channel condition and long-term resource budgets [12–14]. These decisions are tightly coupled over time through the average energy and delay constraints, and the semantic utility associated with each action is generally hard to characterize in a tractable closed form. Hence, classical approaches such as per-slot convex optimization or dynamic programming become either inapplicable or computationally prohibitive in realistic settings [15]. These challenges motivate reinforcement learning (RL)-based learning methods [16–18]. However, it is inefficient to apply a single flat RL policy to the full hybrid action space. The semantic mode and semantic granularity are discrete strategic decisions, whereas transmit power is a continuous tactical decision. This structural heterogeneity motivates a hierarchical learning architecture, in which semantic adaptation and physical-layer adaptation are optimized at different decision levels.

Motivated by the above observations, this paper proposes a hierarchical deep reinforcement learning framework for cross-modal task-oriented semantic communication in low-altitude UAV networks. The UAV first extracts a semantic representation from multiple sensing modalities through a trainable semantic encoder. It then decides which modality combination and semantic granularity should be used in the current slot, and subsequently determines the transmit power for the semantic packet. The overall objective is to maximize the long-term average semantic utility under average energy and delay constraints.

The main contributions are summarized as follows.

- We develop goal-oriented cross-modal semantic communications for a low-altitude UAV uplink system
  , with joint consideration of cross-modal semantic selection, packet granularity, channel-dependent transmission, and long-term energy-delay budgets.
- A task-oriented semantic encoder is designed for cross-modal sensing, which enables semantic packets at different granularities to preserve mission relevance.
- The online semantic communication problem is formulated as a MDP, and a novel hierarchical primal-dual DRL algorithm is developed. The high-level controller is trained by using recurrent proximal policy optimization (PPO) for discrete semantic

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decisions, while the low-level controller is trained using SAC for continuous power control. Dual variables are introduced to enforce long-term average energy and delay constraints.

- Extended simulation results show that the proposed hierarchical DRL-based scheme outperforms several baselines in terms of semantic utility, energy consumption and delay.

2 Related Work

2.1 Semantic Communications

Recent research has considerably broadened the scope of semantic communication from end-to-end representation learning to adaptive, task-aware, and cross-layer system design. Early post-2024 studies investigated semantic transmission under diverse signal modalities and communication settings, including interference-resilient text semantic communication [19], broadcast semantic communication for heterogeneous tasks [20], adaptive modulation and retransmission for semantic links [21], speech semantic communication [22], and task-oriented semantic compression with joint resource adaptation [23]. These works demonstrated that semantic transmission can be made modality-aware, utility-aware, and channel-adaptive, rather than being restricted to fixed encoder-decoder pipelines. Beyond such point-to-point semantic designs, several studies incorporated physical-layer mechanisms and fine-grained task relevance into the semantic stack, such as semantic communication empowered physical-layer network coding [24], feature-importance-aware task-oriented semantic transmission [25], and semantic communication with adaptive channel feedback [9]. Moreover, recent works have further pushed semantic communication toward secure, multimodal, and AI-native designs. Representative examples include IRS-enhanced secure semantic communication with cross-layer context-aware control [26, 27], cross-modal semantic communication built on visual language models [28], and integrated distributed semantic communication with over-the-air computation for cooperative spectrum sensing [29]. In parallel, generative-AI-enabled semantic communication has gained strong momentum, including incentive design in RSMA semantic networks [30], semantic successive refinement [31], robust semantic transmission with asymmetric architectures [32], and OFDM-based digital semantic communication with a learnable channel estimator [33]. Collectively, these works have significantly improved semantic modeling, robustness, and transmission efficiency. However, most existing designs still optimize a pre-specified semantic architecture or a fixed semantic communication task. Limited attention has been paid to online semantic adaptation itself, namely, learning when to switch semantic modality combinations and semantic granularities according to stochastic mission context, channel evolution, and long-term communication budgets.

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2.2 Low-Altitude UAV Networks

In UAV and low-altitude networks, semantic communication has recently been explored from the perspectives of semantic scheduling, semantic-aware perception, cooperative control, and low-altitude service support. Initial works focused on UAVassisted semantic communication in resource-constrained environments, including intelligent semantic resource management under jamming in UAV-enabled MEC [34] and QoE-driven reliable semantic scheduling in UAV-assisted MEC [35]. Subsequent studies moved beyond generic scheduling and incorporated aerial perception semantics more explicitly. Examples include knowledge-graph-enabled UAV cognitive semantic communication for robust object detection [36] and resource allocation for multimodal semantic communication in UAV collaborative networks [37]. These works confirm that semantic communication is particularly attractive for aerial platforms because raw sensory data are expensive to deliver, whereas task-relevant semantics can often be represented much more compactly. Moreover, some work has considered semantic communication in more structured low-altitude and multi-UAV settings. Tang *et al.* [38] investigated decentralized semantic communication and cooperative tracking control for UAV swarms, while Xu *et al.* [39] studied semantic-aware swarm coordination through an incentive-based framework. Recent low-altitude semantic networking papers have also broadened the application space to service migration [40], semantic spectrum map completion [41], remote 3D scene reconstruction [42], lowaltitude intelligent networking [43], aerial-maritime semantic networking [44], and secure relay-assisted semantic communication in low-altitude intelligent networks [45].

2.3 Reinforcement Learning for Wireless Resource Allocation

Reinforcement learning has become a powerful methodology for wireless resource allocation when the control problem is stochastic, temporally coupled, and analytically intractable. Recent transaction papers have applied RL to distributed resource allocation [46], multidimensional wireless-computing resource management [47], secure resource allocation in wireless MEC [48], and dynamic resource allocation with client scheduling in hierarchical federated learning [49]. Safe and energy-aware learning has also been considered, as exemplified by safe DRL for energy-efficient federated learning in wireless communication networks [50] and DRL-based resource allocation in cloud-native wireless systems [51]. These works have shown that RL is particularly effective when conventional model-based optimization becomes cumbersome due to uncertainty, hybrid action spaces, or long-term coupled constraints. Moreover, several studies have combined RL with semantic or aerial communication decisions. Yu *et al.* [52] developed distributed hierarchical DRL for semantic-aware resource allocation, while Li *et al.* [53] considered mean-field multi-agent reinforcement learning for UAV basestation deployment in semantic communication networks. RL has also been used for UAV-related scheduling and deployment problems, such as multimodal charging vian UAVs [54] and AoI-oriented UAV deployment with covert communication [55], and for improving RL scalability or efficiency in wireless networks, such as diffusionassisted DRL training [56] and multi-agent RL-based channel access

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optimization [57]. Nevertheless, most existing RL-based resource allocation papers assume that the traffic abstraction or semantic representation is already fixed. In contrast, in this paper, the problem addressed requires the controller to learn what semantic packet should be transmitted, in terms of modality combination and semantic granularity, before learning how much power should be allocated. This semantic-first, transmission-second coupling motivates a hierarchical RL architecture.

3 System Model

3.1 Cross-Modal Semantic Communication Model

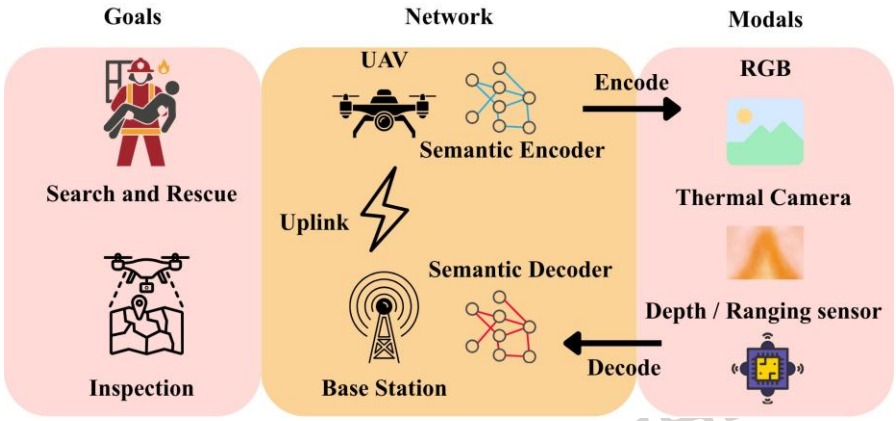


Fig. 1: The considered cross-modal semantic communications in a low-altitude UAV network.

We consider a time-slotted low-altitude UAV communication system, where a single UAV transmits mission-relevant semantic information to a ground station through an uplink wireless channel, as shown in Figure 1. The UAV is equipped with K heterogeneous sensing modalities indexed by K = {1,2,...,K}, which may include, for example, RGB imaging, thermal sensing, and depth or ranging perception. In each slot t, the UAV acquires a set of multimodal observations $\mathbf{x}^t = \{x_t^{(1)}, x_t^{(2)}, \ldots, x_t^{(K)}\}$.

Different from conventional bit-level transmission, the UAV does not directly forward the raw sensing data to the receiver. Instead, it first extracts task-relevant semantic features from the observed multimodal signals, generates a compact semantic packet, and then delivers this packet to the ground station. Upon reception, the ground station decodes the semantic packet and performs the corresponding task interpretation. This semantic-first transmission mechanism is particularly suitable for low-altitude UAV systems, where the sensing payload is typically large, whereas the mission-relevant information required for downstream decision making is often highly structured and substantially more compact.

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To characterize the flexibility of cross-modal semantic generation, we define a finite semantic mode set $\mathrm{M} = \{1,2,\ldots,M\}$, where each semantic mode $m \in \mathrm{M}$ corresponds to a subset of active sensing modalities $\mathrm{S}(m) \subseteq \mathrm{K}$. In other words, the semantic mode specifies which sensing sources are activated and jointly involved in semantic extraction during the current slot. For instance, one mode may rely solely on RGB sensing, another may activate only thermal sensing, while a third mode may jointly exploit multiple modalities to improve semantic reliability under challenging environments.

In addition to modality selection, the UAV also determines the semantic granularity of the transmitted packet. Let $\mathrm{G} = \{1,2,\ldots,G\}$ denote the set of available semantic granularity levels. For each $g \in \mathrm{G}$, let $b_g \in \mathrm{R}+$ denote the corresponding semantic packet size, with $b_1 < b_2 < \cdots < b_G$. A smaller granularity index corresponds to a more compact semantic representation with fewer semantic tokens and lower transmission overhead, whereas a larger granularity index provides finer task-oriented semantic details at the price of a larger payload. Therefore, the semantic granularity captures the fundamental tradeoff between semantic richness and communication cost.

Accordingly, the high-level semantic decision of the UAV in slot t is represented by

$$a_t^H = (m_t, g_t) \in \mathcal{M} \times \mathcal{G}, \tag{1}$$

where m_t determines the active modality combination and g_t specifies the semantic granularity level. This high-level action determines both the semantic content to be generated and the packet size to be transmitted, and thus directly affects the subsequent communication cost and task utility.

3.2 Wireless Uplink Model

Given the semantic action a_t^H, the generated semantic packet is transmitted over an uplink wireless channel. Let h_t denote the channel power gain in slot t, and let $p_t \in [0,P_{\max}]$ denote the UAV transmit power, where $P_{\max}$ is the maximum allowable transmit power. We assume a system bandwidth of B Hz and additive white Gaussian noise with power σ^2. Under these assumptions, the achievable uplink transmission rate in slot t is given by

$$R_t = B \log_2 \left(1 + \frac{p_t h_t}{\sigma^2}\right). \tag{2}$$

It is worth noting that, under the considered architecture, the transmit power does not determine the semantic content itself, but rather controls the efficiency with which the selected semantic packet can be delivered to the ground station. Therefore, once the semantic action (m_t,g_t) is fixed, the remaining communication-level adaptation is fully characterized by the power-control action $a_t^L = p_t$. This naturally leads to a two-level control structure: the high-level action determines what semantic information should be transmitted, whereas the low-level action determines how the selected semantic packet should be conveyed over the wireless channel.

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3.3 Delay Model

The end-to-end latency in each slot consists of two major components: the semantic processing delay required to activate the selected sensing modalities and generate the semantic packet, and the wireless transmission delay required to deliver the packet to the ground station. To keep the model analytically tractable while preserving the key coupling between semantic generation and radio transmission, we adopt the following aggregate delay model:

$$D_t = d_{m_t} + \frac{b_{g_t}}{R_t}, \tag{3}$$

where d_{m_t} denotes the fixed semantic processing delay associated with semantic mode m_t. Specifically, d_{m_t} captures, in an aggregated form, the latency incurred by sensing activation, modality fusion, and onboard semantic extraction under the selected modality combination. The second term, b_{g_t}/R_t, represents the transmission delay of the semantic packet with size b_{g_t} over the uplink channel.

The model in (3) highlights the intrinsic interaction between semantic adaptation and wireless control. On the one hand, choosing a richer modality combination or a finer semantic granularity generally improves semantic fidelity, but also increases the processing burden and packet size. On the other hand, a higher transmit power improves the achievable rate and shortens the transmission delay, at the expense of increased energy consumption. Hence, the delay performance is jointly determined by semantic-level and communication-level decisions.

To ensure real-time operation, the per-slot delay is required to satisfy $D_t \le D_{\max}$, where $D_{\max}$ denotes the maximum tolerable latency for one slot. This constraint guarantees that the selected semantic packet can be generated and delivered within the prescribed timing requirement of the considered UAV mission.

3.4 Energy Model

In addition to delay, energy expenditure is another key performance metric in lowaltitude UAV communications, since the onboard battery budget directly limits sensing, semantic processing, and wireless transmission capabilities. Under the considered semantic communication architecture, the per-slot energy consumption consists of two major components: the semantic processing energy associated with the selected modality combination, and the wireless transmission energy required to deliver the generated semantic packet.

To maintain analytical tractability while preserving the dominant coupling between semantic adaptation and radio transmission, we model the total energy consumption in slot t as

$$E_t = e_{m_t} + p_t \frac{b_{g_t}}{R_t}, \tag{4}$$

where e_{m_t} denotes the fixed semantic processing energy associated with semantic mode m_t. Specifically, e_{m_t} captures, in an aggregated manner, the energy cost incurred by sensing activation, modality fusion, and onboard semantic encoding under the selected

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modality combination. The second term, $p_t b_{g_t}/R_t$, represents the wireless transmission energy required to deliver a semantic packet of size b_{g_t} at transmission rate R_t.

The model in (4) is intentionally simplified, yet it retains the essential dependence of energy expenditure on both the semantic decision and the radio control action. In particular, selecting a richer modality combination or a finer semantic granularity generally increases the semantic processing burden and packet size, whereas increasing the transmit power accelerates packet delivery but also raises transmission energy consumption. Therefore, similar to the delay model in (3), the energy cost is jointly determined by the high-level semantic action and the low-level power-control action.

3.5 Semantic Utility

While delay and energy characterize the communication cost of semantic transmission, they do not directly reflect the mission value of the transmitted semantic packet. To capture the usefulness of the conveyed semantics for the underlying UAV task, we introduce a semantic utility metric that measures how informative the selected semantic representation is under the current sensing condition and mission state.

Let $u_t \in [0,1]$ denote the task urgency in slot t, where a larger value indicates that successful semantic delivery is more critical to the mission. In addition, let $\mathbf{q}_t = [q_t^{(1)}, \ldots, q_t^{(K)}]$ denote the modality-quality vector, where $q_t^{(k)} \in [0,1]$ quantifies the instantaneous reliability of modality k. This quality indicator reflects the fact that the effectiveness of a semantic mode depends not only on which modalities are selected, but also on how informative these modalities are under the current environment. In practical autonomous operations, the instantaneous modality-quality vector q_t is estimated online without ground-truth labels by extracting lightweight, unsupervised statistical descriptors from raw sensor streams during the preprocessing phase.

For a given semantic action (m_t, g_t), let the semantic utility predictor output

$$s_t = \Phi_\omega(m_t, g_t, \mathbf{q}_t), \qquad 0 \le s_t \le 1, \tag{5}$$

where $\Phi_\omega(\cdot)$ is the pretrained semantic encoder. The quantity s_t can be interpreted as the normalized semantic effectiveness of the selected modality combination and semantic granularity under the current modality-quality state. In particular, the predictor $\Phi_\omega(\cdot)$ is grounded in the downstream task performance achieved on an offline validation dataset. It maps the discrete semantic mode m_t, semantic granularity g_t, and the continuous modality-quality vector q_t to an expected task success metric. To ensure online execution efficiency, $\Phi_\omega(\cdot)$ is instantiated as a lightweight multi-layer perceptron regression head.

Based on s_t, we define the instantaneous semantic utility in slot t as

$$U_t = u_t s_t. \tag{6}$$

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This definition explicitly couples semantic quality with mission urgency. In particular, the same semantic packet may yield different utility values under different mission states: when the task urgency is high, transmitting informative semantics becomes significantly more valuable; when the urgency is low, the benefit of delivering the same packet is correspondingly reduced. Therefore, (6) provides a compact abstraction for quantifying the task-oriented benefit of semantic communication, and it serves as the reward basis for the subsequent constrained control problem.

4 Task-Oriented Semantic Coding Framework

To support cross-modal task-oriented semantic transmission, the UAV is equipped with an onboard semantic encoder that converts heterogeneous sensory observations into compact semantic packets with controllable granularity. The encoder is designed to preserve mission-relevant information while enabling flexible adaptation to different modality combinations and communication budgets. To this end, the overall semantic encoding pipeline consists of four components, namely, modality-specific feature extraction, cross-modal semantic fusion, multi-granularity semantic token selection, and semantic packet generation. The resulting semantic packet is finally interpreted by a task head at the receiver.

4.1 Modality-Specific Encoders

For each sensing modality $k \in \mathrm{K}$, the UAV employs a modality-specific encoder $E_k(\cdot;\theta_k)$ to extract latent semantic features from the raw observation $x_t^{(k)}$. Specifically, the encoded representation of modality k in slot t is given by

$$\mathbf{F}_t^{(k)} = E_k(x_t^{(k)};\theta_k) \in \mathbb{R}^{N_k \times d}, \tag{7}$$

where N_k denotes the number of latent tokens produced by the kth encoder and d is the embedding dimension. Here, each row of $\mathbf{F}_t^{(k)}$ corresponds to one latent semantic token associated with modality k.

To obtain a compact modality-level representation for subsequent fusion, we further compute the pooled embedding

$$\bar{\mathbf{f}}_t^{(k)} = \frac{1}{N_k}\sum_{n=1}^{N_k} \mathbf{f}_{t,n}^{(k)} \in \mathbb{R}^d, \tag{8}$$

where $\mathbf{f}_{t,n}^{(k)}$ denotes the nth latent token in $\mathbf{F}_t^{(k)}$. The pooled feature $\bar{\mathbf{f}}_t^{(k)}$ serves as a compact semantic descriptor of modality k and will be used to determine the relative importance of each active modality under the current sensing condition.

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4.2 Cross-Modal Fusion

Given the semantic mode m_t, only the modalities in the active set $S(m_t)$ participate in semantic generation. Since different sensing modalities may exhibit substantially different reliability under varying environments, the semantic encoder should adaptively emphasize the more informative modalities rather than treating all active modalities equally. To this end, we adopt an attention-based fusion mechanism to combine modality-specific semantic features.

Specifically, for each active modality $k \in S(m_t)$, an attention coefficient is computed as

$$\alpha_t^{(k)} = \frac{\exp\left(\mathbf{w}^\top \tanh\left(\mathbf{W}\bar{\mathbf{f}}_t^{(k)}\right)\right)}{\sum_{j\in\mathcal{S}(m_t)} \exp\left(\mathbf{w}^\top \tanh\left(\mathbf{W}\bar{\mathbf{f}}_t^{(j)}\right)\right)}, \quad k \in \mathcal{S}(m_t) \tag{9}$$

where $\mathbf{W}$ and $\mathbf{w}$ are trainable parameters. The coefficient $\alpha_t^{(k)}$ characterizes the relative contribution of modality k to the fused semantic representation.

Based on these attention coefficients, the aggregated cross-modal semantic embedding is given by

$$\mathbf{z}^t = \sum_{k\in S(m_t)} \alpha_t^{(k)} \bar{\mathbf{f}}_t^{(k)} \in \mathbb{R}^d. \tag{10}$$

The fused embedding $\mathbf{z}_t$ provides a unified semantic description of the active multimodal observations and forms the basis for subsequent semantic token generation and granularity adaptation.

4.3 Multi-Granularity Semantic Token Selection

To support adaptive semantic transmission under different communication budgets, the encoder further transforms the fused embedding $\mathbf{z}_t$ into a set of candidate semantic tokens and then selects a granularity-dependent subset of them for transmission. Specifically, a token generator $P_\rho(\cdot)$ first maps $\mathbf{z}_t$ to a semantic token matrix

$$\mathbf{Z}_t = P_\rho(\mathbf{z}_t) \in \mathrm{R}^{L\times d}, \tag{11}$$

where L denotes the number of candidate semantic tokens. Each row of $\mathbf{Z}_t$ corresponds to one candidate token that may be retained or suppressed depending on the selected semantic granularity.

To quantify the relative importance of these candidate tokens, we introduce a semantic importance scorer $G_\psi(\cdot)$ and obtain

$$\mathbf{r}_t = G_\psi(\mathbf{Z}_t) \in \mathrm{R}^L, \tag{12}$$

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where the ℓth entry of $\mathbf{r}_t$ measures the relevance of the ℓth token to the downstream task.

For semantic granularity level g, let K_g denote the number of retained semantic tokens, with $K_1 < K_2 < \cdots < K_G$. To maintain differentiability during training, we use a relaxed top-K_g operator to generate a soft token-selection mask

$$\mathbf{m}_t = \text{TopKRelax}(\mathbf{r}_t, K_g) \in [0,1]^L. \tag{13}$$

The granularity-specific semantic representation is then formed as

$$\mathbf{Z}_t^{(g)} = \text{diag}(\mathbf{m}_t^{(g)})\, \mathbf{Z}_t \in \mathbb{R}^{L \times d}. \tag{14}$$

The construction in (14) enables the semantic encoder to generate packets with different semantic resolutions. A smaller value of K_g leads to a more compact representation and lower transmission overhead, whereas a larger value of K_g preserves richer semantic details for the downstream task.

4.4 Quantization and Semantic Packet Formation

The selected semantic token matrix $\mathbf{Z}_t^{(g)}$ is subsequently converted into a transmissionready semantic packet through a quantization module. Specifically, we denote the quantized semantic packet by

$$\mathbf{c}_t^{(g)} = Q(\mathbf{Z}_t^{(g)}), \tag{15}$$

where $Q(\cdot)$ represents the semantic quantizer and $\mathbf{c}_t^{(g)}$ is the semantic packet transmitted over the wireless uplink.

Let $\ell(\mathbf{c}_t^{(g)})$ denote the resulting packet length in bits. Then the expected packet size associated with granularity level g is approximated by the inline relation $b_g \approx \mathrm{E}[\ell(\mathbf{c}_t^{(g)})]$. Therefore, the semantic granularity directly controls the communication payload through the number of retained semantic tokens and the corresponding quantized representation.

4.5 Task Head

At the receiver, the received semantic packet is decoded and fed to a task head to recover task-oriented outputs. Denote the task head by $H_\varphi(\cdot)$. Its output corresponding to granularity level g is given by

$$\hat{\mathbf{y}}_t^{(g)} = H_\varphi(\mathbf{c}_t^{(g)}). \tag{16}$$

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To keep the formulation general while retaining sufficient task specificity, we consider a task head composed of one classification branch and one regression branch, namely,

$$\hat{\mathbf{y}}_t^{(g)} = \left(\hat{\mathbf{y}}_{t,\mathrm{cls}}^{(g)}, \hat{\mathbf{y}}_{t,\mathrm{reg}}^{(g)}\right), \tag{17}$$

where $\hat{\mathbf{y}}_{t,}^{(g_{\mathrm{cls}})}$ denotes the classification output and $\hat{\mathbf{y}}_{t,}^{(g_{\mathrm{reg}})}$ denotes the regression output. Depending on the specific UAV mission, the regression branch may represent location, confidence score, motion state, or other continuous task-dependent quantities. The task head output will be used in the next subsection to construct the semantic training objective.

5 Semantic Encoder Training Objective

The semantic encoder introduced in the previous section is trained offline to generate compact yet task-relevant semantic packets under different modality combinations and semantic granularities. To ensure that the learned semantic representation is informative, cross-modally consistent, granularity-adaptive, and communication-efficient, we construct the overall training objective from five components: a task loss, a crossmodal alignment loss, a multi-granularity consistency loss, a quantization loss, and a rate regularization term.

5.1 Task Loss

The primary purpose of the semantic encoder is to preserve information that is directly relevant to the downstream UAV mission. Accordingly, the basic training objective is a task-oriented loss defined on the output of the task head. Since the considered task head contains both a classification branch and a regression branch, the task loss is given by

$$L_{\mathrm{task}} = L_{\mathrm{cls}} + \lambda_{\mathrm{reg}} L_{\mathrm{reg}}, \tag{18}$$

where λ_{reg} is a nonnegative weighting coefficient. The classification loss is defined as

$$L_{\mathrm{cls}} = -\sum_{c=1}^{C} y_{t,c} \log \hat{y}_{t,c(g)}, \tag{19}$$

where $y_{t,c}$ is the ground-truth label indicator for class c, and $\hat{y}_{t,c}^{(g)}$ denotes the predicted probability of class c from the classification branch at semantic granularity level g. The regression loss is given by

$$\mathcal{L}_{\mathrm{reg}} = \left\| \hat{\mathbf{y}}_{t,\mathrm{reg}}^{(g)} - \mathbf{y}_{t,\mathrm{reg}} \right\|_1, \tag{20}$$

where $\mathbf{y}_{t,\mathrm{reg}}$ is the ground-truth regression target. The task loss in (18) ensures that the transmitted semantic packet remains directly useful for the underlying UAV mission rather than merely preserving generic latent features.

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5.2 Cross-Modal Alignment Loss

Since different sensing modalities may observe the same physical scene from complementary perspectives, their semantic representations should remain consistent at the latent level. To encourage semantic agreement across active modalities, we introduce a contrastive cross-modal alignment loss.

For two active modalities *i* and *j*, the normalized pooled embeddings are defined as

$$\tilde{\mathbf{f}}_t^{(i)} = \frac{\bar{\mathbf{f}}_t^{(i)}}{\|\bar{\mathbf{f}}_t^{(i)}\|_2}, \qquad \tilde{\mathbf{f}}_t^{(j)} = \frac{\bar{\mathbf{f}}_t^{(j)}}{\|\bar{\mathbf{f}}_t^{(j)}\|_2}, \tag{21}$$

where $\bar{\mathbf{f}}_t^{(i)}$ and $\bar{\mathbf{f}}_t^{(j)}$ are the modality-level pooled semantic embeddings defined in (8). We then adopt an InfoNCE-style objective:

$$L_{align} = -\log \frac{\exp(\mathrm{sim}(\tilde{f}_t^{(i)}, \tilde{f}_t^{(j)})/\tau)}{\exp(\mathrm{sim}(\tilde{f}_t^{(i)}, \tilde{f}_t^{(j)})/\tau) + \sum_{n=1}^{N_{neg}} \exp(\mathrm{sim}(\tilde{f}_t^{(i)}, \tilde{f}_{-n})/\tau)} \tag{22}$$

where $\mathrm{sim}(a,b) = a^\top b$ denotes the cosine-style similarity between normalized embeddings, τ is a temperature parameter, and $\tilde{f}_{-n}$ are negative samples. When more than two modalities are activated, the alignment loss is averaged over all positive modality pairs. This term encourages semantically corresponding observations from different modalities to lie close in the latent space, thereby improving the robustness of cross-modal semantic fusion.

5.3 Multi-Granularity Consistency Loss

A key requirement of task-oriented semantic communication is that coarse-grained semantic packets should still preserve the task-relevant information contained in finegrained semantic packets. In other words, reducing the semantic granularity should decrease semantic richness, but should not destroy the essential information needed for downstream inference. To enforce this property, we introduce a multi-granularity consistency loss.

Let *G* denote the finest semantic granularity level. We treat the prediction produced at granularity *G* as a semantic teacher and regularize all coarser granularities to produce consistent task outputs. The resulting consistency loss is defined as

$$\mathcal{L}_{\mathrm{mg}} = \sum_{g=1}^{G-1} \mathrm{KL}\Big(\mathrm{softmax}\big(\hat{\mathbf{y}}_{t,\mathrm{cls}}^{(G)}/T\big) \Big\| \mathrm{softmax}\big(\hat{\mathbf{y}}_{t,\mathrm{cls}}^{(g)}/T\big)\Big) + \lambda_{\mathrm{mg}} \sum_{g=1}^{G-1} \Big\|\hat{\mathbf{y}}_{t,\mathrm{reg}}^{(G)} - \hat{\mathbf{y}}_{t,\mathrm{reg}}^{(g)}\Big\|_2^2, \tag{23}$$

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where T is the distillation temperature and λ_{mg} is the weighting coefficient of the regression consistency term. The first term enforces consistency among classification outputs across granularity levels, while the second term regularizes the regression outputs. This design encourages the encoder to retain mission-critical information even when only a compact subset of semantic tokens is transmitted.

5.4 Quantization Loss

After semantic token selection, the granularity-specific token matrix $\mathbf{Z}_t^{(g)}$ is quantized into the semantic packet $\mathbf{c}_t^{(g)}$. To reduce the mismatch between the continuous semantic representation and its quantized codeword, we employ a vector-quantization loss:

$$\mathcal{L}_{\mathrm{vq}} = \left\| \mathrm{sg}\left[\mathbf{Z}_t^{(g)}\right] - \mathbf{e}\left(\mathbf{c}_t^{(g)}\right) \right\|_F^2 + \beta_{\mathrm{vq}} \left\| \mathbf{Z}_t^{(g)} - \mathrm{sg}\left[\mathbf{e}\left(\mathbf{c}_t^{(g)}\right)\right] \right\|_F^2, \tag{24}$$

where $\mathbf{e}(\mathbf{c}_t^{(g)})$ denotes the selected codeword embedding, sg[·] is the stop-gradient operator, and $\|\cdot\|_F$ is the Frobenius norm. The first term aligns the codebook embedding with the encoder output, while the second term, weighted by β_{vq}, commits the encoder output to the selected codeword. This loss is essential for producing stable and compact semantic packets that can be reliably transmitted over the wireless channel.

5.5 Rate Regularization

To explicitly discourage verbose semantic packets, we further impose a rate regularization term based on the code length of the quantized semantic packet:

$$\mathcal{L}_{\mathrm{rate}} = \mathbb{E}\left[-\log p(\mathbf{c}_t^{(g)}) \right], \tag{25}$$

where $p(\mathbf{c}_t^{(g)})$ denotes the probability model associated with the semantic codeword sequence. This term serves as an estimate of the expected coding length and encourages communication-efficient semantic representations.

5.6 Overall Semantic Encoder Objective

Combining the above components, the overall semantic encoder training objective is given by

$$L_{\mathrm{sem}} = \lambda_1 L_{\mathrm{task}} + \lambda_2 L_{\mathrm{align}} + \lambda_3 L_{\mathrm{mg}} + \lambda_4 L_{\mathrm{vq}} + \lambda_5 L_{\mathrm{rate}}, \tag{26}$$

where $\lambda_1,\ldots,\lambda_5$ are nonnegative weighting coefficients balancing task fidelity, crossmodal consistency, granularity robustness, quantization quality, and communication compactness.

After offline training, the semantic encoder and task head provide two quantities needed by the online controller: the semantic packet sizes b_g associated with different granularity levels, and the semantic utility score $s_t = \Phi_\omega(m_t, g_t, \mathbf{q}_t)$ used in the online decision process.

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6 Energy- and Delay-Constrained Semantic Control Problem

Based on the pretrained semantic encoder, the online controller determines the semantic action and the transmission power in each slot so as to maximize the long-term mission-oriented utility under communication resource constraints. Since the semantic utility, delay, and energy consumption are all affected by both semantic-level and communication-level decisions, the resulting problem is naturally formulated as a constrained sequential decision process.

Let the system state at slot t be defined as the inline expression $s_t = \left[h_t, \mathbf{q}_t, u_t, \lambda_t^E, \lambda_t^D\right]$, where h_t is the uplink channel gain, $\mathbf{q}_t$ is the modality-quality vector, u_t is the task urgency, and λ_t^E and λ_t^D are dual variables associated with the long-term energy and delay constraints, respectively. The high-level action is the semantic decision $a_t^H = (m_t, g_t)$, while the low-level action is the transmit power $a_t^L = p_t$. Let $\pi = (\pi_H, \pi_L)$ denote the overall hierarchical control policy.

The long-term objective is to maximize the time-average semantic utility, namely,

$$\max_{\pi} \quad \liminf_{T\to\infty} \frac{1}{T}\mathbb{E}_{\pi}\left[\sum_{t=1}^{T} U_t\right], \tag{27}$$

subject to the average energy constraint

$$\limsup_{T\to\infty} \frac{1}{T}\mathbb{E}_{\pi}\left[\sum_{t=1}^{T} E_t\right] \le \bar{E}, \tag{28}$$

the average delay constraint

$$\limsup_{T\to\infty} \frac{1}{T}\mathbb{E}_{\pi}\left[\sum_{t=1}^{T} D_t\right] \le \bar{D}, \tag{29}$$

and the instantaneous feasibility constraints

$$D_t \le D_{\max}, \qquad p_t \in [0, P_{\max}]. \tag{30}$$

The optimization problem in (27)–(30) defines a Markov decision process (MDP), where the long-term average constraints couple decisions across time and prevent a simple slot-wise decomposition. For a fixed semantic action (m_t, g_t), the slot-level delay constraint $D_t \le D_{\max}$ imposes a minimum power requirement. Specifically, substituting the delay model in (3) and the achievable rate in (2) leads to the minimum feasible transmit power

$$p_{t,\min} = \frac{\sigma^2}{h_t}\left(2^{\frac{b_{g_t}}{B(D_{\max}-d_{m_t})}} - 1\right), \tag{31}$$

provided that $D_{\max} > d_{m_t}$. Accordingly, the feasible transmission-power region is given by the inline expression $p_t \in [p_{t,\min}, P_{\max}]$.

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Lemma 1. *For a given semantic action $a^H_t = (m_t, g_t)$ and channel gain h_t, a necessary condition for the per-slot delay constraint $D_t \le D_{\max}$ to be feasible is that the maximum allowable delay strictly exceeds the onboard processing delay, denoting $D_{\max} > d_{m_t}$. Under this condition, the minimum feasible transmit power is uniquely determined by* (31).

Proof. According to the transmission and processing framework, the total slot-level delay is expressed as $D_t = d_{m_t} + \frac{b_{g_t}}{R_t}$, where d_{m_t} represents the processing delay of the selected modality combination and $\frac{b_{g_t}}{R_t}$ represents the wireless transmission delay. Consequently, the per-slot delay constraint $D_t \le D_{\max}$ can be explicitly written as

$$d_{m_t} + \frac{b_{g_t}}{R_t} \le D_{\max}. \tag{32}$$

Since the transmission rate R_t must be strictly positive, (32) can only be satisfied if $D_{\max} - d_{m_t} > 0$. Rearranging the terms in (32) under this condition isolates the minimum required transmission rate as follows:

$$R_t \ge \frac{b_{g_t}}{D_{\max} - d_{m_t}}. \tag{33}$$

By substituting the physical-layer Shannon capacity formula $R_t = B \log_2 \left(1 + \frac{p_t h_t}{\sigma^2}\right)$ into (33), we obtain the corresponding inequality regarding the communication parameters:

$$B \log_2 \left(1 + \frac{p_t h_t}{\sigma^2}\right) \ge \frac{b_{g_t}}{D_{\max} - d_{m_t}}. \tag{34}$$

Dividing both sides of (34) by the transmission bandwidth B and applying the monotonically increasing exponential function with base 2 to both sides leads to

$$1 + \frac{p_t h_t}{\sigma^2} \ge 2^{\frac{b_{g_t}}{B(D_{\max} - d_{m_t})}}. \tag{35}$$

Finally, subtracting 1 from both sides of (35) and multiplying the entire expression by $\frac{\sigma^2}{h_t}$ isolates the transmit power variable:

$$p_t \ge \frac{\sigma^2}{h_t} \left(2^{\frac{b_{g_t}}{B(D_{\max} - d_{m_t})}} - 1\right). \tag{36}$$

The boundary condition where the equality holds defines the lower bound of the feasible transmit power, which directly leads to the definition of $p_{t,\min}$ in (31). This completes the proof. □

The Lemma 1 demonstrates that high-level semantic decisions directly determine the mathematical boundaries of the low-level communication action space. In particular, selecting a finer semantic granularity that requires a larger packet size b_{g_t}, or choosing a modality combination that incurs a higher onboard computational delay d_{m_t},

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strictly scales and shifts the minimum required transmit power $p_{t,\min}$. As a result, the physical-layer action space is dynamically bounded within the restricted interval $[p_{t,\min}, P_{\max}]$, and a valid transmit power allocation exists if and only if $p_{t,\min} \le P_{\max}$ is satisfied. This cross-layer dependency mathematically underscores why semantic abstraction and physical resource allocation cannot be optimized in isolation, thereby justifying the necessity of the proposed joint hierarchical deep reinforcement learning framework.

Proposition 1. *For fixed* (m_t, g_t) *and* h_t*, the transmission energy*

$$E_t^{\mathrm{tx}} = p_t \frac{b_{g_t}}{B \log_2\left(1 + \frac{p_t h_t}{\sigma^2}\right)} \tag{37}$$

is strictly increasing in p_t *over the feasible interval* $[p_{t,\min}, P_{\max}]$.

Proof. Let $a = h_t/\sigma^2$ and $b = b_{g_t}/B$. Then (37) can be rewritten as

$$E_t^{\mathrm{tx}} = \frac{b\, p_t}{\log_2(1 + a p_t)} = \frac{b \ln 2\, p_t}{\ln(1 + a p_t)}. \tag{38}$$

Differentiating with respect to p_t leads to

$$\frac{dE_t^{\mathrm{tx}}}{dp_t} = b \cdot \ln 2 \cdot \frac{\ln(1 + a p_t) - \frac{a p_t}{1 + a p_t}}{[\ln(1 + a p_t)]^2}. \tag{39}$$

Define $\phi(x) = \ln(1 + x) - \frac{x}{1+x}$ for $x > 0$. Then $\phi(0) = 0$ and

$$\phi'(x) = \frac{x}{(1 + x)^2} > 0, \tag{40}$$

which implies $\phi(x) > 0$ for all $x > 0$. Therefore, $dE_t^{\mathrm{tx}}/dp_t > 0$, and the proof is complete.

□

Proposition 1 indicates that, once the semantic action is fixed, increasing the transmit power always increases the transmission energy, even though it reduces the packet delivery delay. This observation further highlights the fundamental delay–energy tradeoff underlying the considered semantic control problem.

7 Method

The CMDP formulated in the previous section involves hybrid decision variables, unknown state-transition statistics, and long-term average constraints on both energy consumption and delay. Although the problem admits a clear semantic-communication structure, obtaining an optimal control policy via conventional dynamic programming is computationally prohibitive due to the enlarged state-action space and the lack of an

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explicit closed-form characterization of the long-term semantic utility. To address these difficulties, we develop a hierarchical primal-dual DRL framework, in which the high-level controller determines the semantic action and the low-level controller adjusts the transmit power, while the dual variables are updated online to enforce the average energy and delay constraints.

7.1 Lagrangian Reformulation

To handle the long-term average constraints in (28) and (29), we introduce two nonnegative dual variables, namely, $\lambda_t^E \geq 0$ and $\lambda_t^D \geq 0$, associated with the energy and delay constraints, respectively. Based on these dual variables, the constrained control problem is transformed into an unconstrained sequential decision problem with the instantaneous Lagrangian reward

$$\tilde{r}_t = U_t - \lambda^E{}_t (E_t - \bar{E}) - \lambda_t^D (D_t - \bar{D}). \tag{41}$$

The reward in (41) explicitly balances semantic utility against resource-constraint violations. In particular, when the instantaneous energy or delay exceeds the prescribed average budget, the corresponding penalty term becomes active and discourages the controller from repeatedly selecting overly expensive actions. In this way, the dual variables serve as adaptive prices that guide the learning process toward budget-feasible policies.

Under this reformulation, the control objective is to maximize the long-term discounted return

$$J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t \tilde{r}_t \right], \tag{42}$$

where $\gamma \in (0,1)$ is the discount factor and $\pi = (\pi_H, \pi_L)$ denotes the overall hierarchical policy. The resulting formulation naturally lends itself to a primal-dual learning procedure, where the policy parameters are updated to maximize (42), and the dual variables are updated according to the observed budget violations.

7.2 High-Level Policy: Recurrent PPO for Semantic Action Selection

The high-level controller is responsible for selecting the semantic action, namely, the modality combination and the semantic granularity. Since the effectiveness of a semantic action depends on the current sensing quality, task urgency, and past environmental evolution, we model the high-level policy as a recurrent stochastic policy. Specifically, the semantic action is sampled as the inline expression $a^H{}_t = (m_t, g_t) \sim \pi_H(\cdot|s_t;\theta_H)$, where θ_H denotes the policy parameters.

The high-level policy is implemented as a categorical recurrent network:

$$\pi_H(a_t^H | s_t; \theta_H) = \text{Cat softmax}\big(f_{\theta_H}(s_t, h_t^r)\big), \tag{43}$$

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where h^r_t is the hidden state of the recurrent unit. The recurrent structure is introduced to capture temporal correlations in the channel and modality-quality evolution, which are difficult to exploit using a purely feedforward policy.

To evaluate the long-term effect of a high-level semantic action, we define the value function

$$V_{\psi_H}(s_t) = \mathbb{E}_{\pi_H,\pi_L}\left[\sum_{l=0}^{\infty}\gamma^l \tilde{r}_{t+l}\middle| s_t\right], \tag{44}$$

where ψ_H denotes the parameters of the value network. Based on the value function, the temporal-difference residual is given by

$$\delta_t = \tilde{r}_t + \gamma V_{\psi_H}(s_{t+1}) - V_{\psi_H}(s_t). \tag{45}$$

To reduce variance while preserving long-term credit assignment, we adopt generalized advantage estimation (GAE), which leads to

$$\hat{A}_t = \sum_{l=0}^{L-1}(\gamma\xi)^l\delta_{t+l}, \tag{46}$$

where $\xi \in [0,1]$ is the GAE parameter and L is the truncation horizon. The resulting advantage estimate is then used in the PPO update.

Define the probability ratio as

$$r_t(\theta_H) = \frac{\pi_H(a_t^H|s_t;\theta_H)}{\pi_H(a_t^H|s_t;\theta_H^{\text{old}})}. \tag{47}$$

The clipped PPO objective is given by

$$\begin{aligned}\mathcal{J}_{\text{PPO}}(\theta_H) = \mathbb{E}\Big[\min\Big(r_t(\theta_H)\hat{A}_t, \text{clip}\big(r_t(\theta_H),1-\epsilon,1+\epsilon\big)\hat{A}_t\Big)\Big] \\ + \beta_H\mathbb{E}\big[\mathcal{H}\big(\pi_H(\cdot|s_t)\big)\big],\end{aligned} \tag{48}$$

where ϵ is the clipping parameter and β_H controls the entropy regularization. The entropy term improves exploration over the discrete semantic action space and helps prevent premature convergence to suboptimal semantic modes.

The value network is updated by minimizing the squared-error loss

$$\mathcal{L}_V(\psi_H) = \mathbb{E}\left[\left(V_{\psi_H}(s_t) - \hat{R}_t\right)^2\right], \tag{49}$$

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where the target return is given inline by $\hat{R}_t = \hat{A}_t + V_{\psi H}(s_t)$. Overall, the recurrent PPO module learns how to adapt the modality combination and semantic granularity so as to maximize the long-term Lagrangian reward.

7.3 Low-Level Policy: SAC for Transmit Power Control

Once the semantic action $a^H_t = (m_t, g_t)$ has been selected, the remaining control variable is the transmit power. Since the power action is continuous and directly affects both the transmission delay and the transmission energy, we employ soft actorcritic (SAC) to learn the low-level policy. Specifically, the low-level controller samples the transmit power according to the inline expression $p_t \sim \pi_L(\cdot|s_t, a^H_t; \theta_L)$, where θ_L denotes the actor parameters.

To ensure that the selected power always satisfies the slot-level delay feasibility requirement, the actor first outputs a normalized auxiliary variable $\rho_t \in [-1,1]$, which is then mapped to the feasible power interval:

$$p_t = p_{t,\min} + \frac{\rho_t + 1}{2}\left(P_{\max} - p_{t,\min}\right). \tag{50}$$

This mapping guarantees that the chosen transmit power automatically lies in the admissible region $[p_{t,\min}, P_{\max}]$, thereby avoiding infeasible exploratory actions during training.

Let $Q_{\phi_i}(s_t, a^H_t, p_t)$, $i \in \{1,2\}$, denote the two critic networks used by SAC. The corresponding soft target is

$$y_t = \tilde{r}_t + \gamma \mathbb{E}_{p_{t+1}\sim\pi_L}\left[\min_{j=1,2} \bar{Q}_{\phi_j}(s_{t+1}, a^H_{t+1}, p_{t+1}) - \alpha \log \pi_L(p_{t+1}|s_{t+1}, a^H_{t+1})\right], \tag{51}$$

where $\bar{Q}_{\phi_j}(\cdot)$ denotes the target critic and α is the entropy temperature. The critic networks are trained by minimizing

$$\mathcal{L}_Q(\phi_i) = \mathbb{E}\left[\left(Q_{\phi_i}(s_t, a^H_t, p_t) - y_t\right)^2\right], \tag{52}$$

for $i \in \{1,2\}$.

The actor is updated by maximizing the entropy-regularized objective, or equivalently by minimizing

$$\mathcal{J}_{\mathrm{SAC}}(\theta_L) = \mathbb{E}\left[\alpha \log \pi_L(p_t|s_t, a^H_t) - \min_{i=1,2} Q_{\phi_i}(s_t, a^H_t, p_t)\right] \tag{53}$$

The first term promotes sufficient exploration over the feasible power space, while the second term drives the actor toward power decisions with high long-term Lagrangian value.

To adapt the exploration-exploitation balance automatically, the entropy temperature is updated via

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$$\mathcal{L}_\alpha = \mathbb{E}\left[-\alpha\left(\log \pi_L(p_t|s_t, a_t^H) + \bar{\mathcal{H}}\right)\right], \tag{54}$$

where $\bar{H}$ is the target entropy. Through this mechanism, the low-level SAC controller learns a budget-aware power-allocation strategy conditioned on both the current system state and the semantic action selected by the high-level controller.

7.4 Dual Variable Updates

The primal policy updates described above optimize the Lagrangian reward for fixed dual variables. To enforce the long-term average energy and delay constraints, the dual variables are updated in the dual ascent direction based on the observed instantaneous budget violations. Specifically, the energy-related dual variable is updated as

$$\lambda_{t+1}^E = \left[\lambda_t^E + \eta_E(E_t - \bar{E})\right]^+, \tag{55}$$

and the delay-related dual variable is updated as

$$\lambda_{t+1}^D = \left[\lambda_t^D + \eta_D(D_t - \bar{D})\right]^+, \tag{56}$$

where η_E and η_D are positive stepsizes and $[x]^+ = \max\{x,0\}$ denotes projection onto the nonnegative orthant.

The updates in (55) and (56) increase the corresponding dual price whenever the realized energy or delay exceeds the prescribed average budget, and reduce the penalty pressure otherwise. Hence, the dual variables act as adaptive budget regulators that progressively steer the learned policy toward energy- and delay-feasible operation. The details of the proposed method is summarized in Algorithm 1.

Algorithm 1 Proposed DRL for Cross-Modal UAV Semantic Communications

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1: Train modality-specific encoders $\{E_k(\cdot;\theta_k)\}_{k\in\mathrm{K}}$, fusion module, token generator $P_\rho(\cdot)$, scorer $G_\psi(\cdot)$, quantizer $Q(\cdot)$, and task head $H_\varphi(\cdot)$ by minimizing (26)

2: Obtain semantic packet sizes $\{b_g\}_{g\in\mathrm{G}}$ and the utility predictor $\Phi_\omega(m,g,\mathbf{q}_t)$

3: Initialize high-level policy parameters θ_H and value network parameters ψ_H

4: Initialize low-level actor parameters θ_L, critic parameters ϕ_1,ϕ_2, and target critic parameters $\bar{\phi}_1,\bar{\phi}_2$, replay buffer D and dual variables λ_0^E, λ_0^D

5: **for** each time slot t **do**

6: Observe state $s_t = [h_t,\mathbf{q}_t,u_t,\lambda^E{}_t,\lambda^D{}_t]$

7: Sample high-level semantic action $a^H{}_t = (m_t,g_t) \sim \pi_H(\cdot|s_t;\theta_H)$

8: Determine the active modality set $\mathrm{S}(m_t)$

9: **for** each active modality $k \in \mathrm{S}(m_t)$ **do**

10: Acquire raw observation $x_t^{(k)}$

11: Extract modality-specific semantic tokens $\mathbf{F}_t^{(k)} = E_k(x_t^{(k)};\theta_k)$ via (7)

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Compute pooled modality embedding $\bar{\mathbf{f}}_t^{(k)}$ via (1)
end for
Compute cross-modal attention weights $\{\alpha_t^{(k)}\}_{k\in\mathcal{S}(m_t)}$ via (9)
Obtain fused semantic embedding $\mathbf{z}_t$ via (10)
Generate candidate semantic token matrix $\mathbf{Z}_t = P_\rho(\mathbf{z}_t)$ via (11)
Compute semantic importance scores $\mathbf{r}_t = G_\psi(\mathbf{Z}_t)$ via (12)
Generate granularity-dependent mask $\mathbf{m}_t^{(g_t)}$ = TopKRelax($\mathbf{r}_t, K_{g_t}$) via (13)
Obtain granularity-specific semantic token matrix $\mathbf{Z}_t^{(g_t)}$ via (14)
Quantize $\mathbf{Z}_t^{(g_t)}$ to the semantic packet $\mathbf{c}_t^{(g_t)} = Q(\mathbf{Z}_t^{(g_t)})$ via (15)
Compute the minimum feasible transmit power $p_{t,\min}$ via (31)
Sample ρ_t from the low-level actor and map it to p_t using (50)
Transmit $\mathbf{c}_t^{(g_t)}$ over the uplink with power p_t
Decode the received semantic packet and obtain task output $\hat{\mathbf{y}}_t^{(g_t)} = H_\varphi(\mathbf{c}_t^{(g_t)})$ via (16)
Compute semantic score $s_t = \Phi_\omega(m_t, g_t, \mathbf{q}_t)$ via (5)
Compute achievable rate R_t via (2)
Compute delay D_t, energy E_t, and utility U_t via (3), (4), and (6)
Form Lagrangian reward $\tilde{r}_t$ via (41)
Observe next state s_{t+1} and store transition $(s_t, a^H_t, p_t, \tilde{r}_t, s_{t+1})$ in D
Update SAC critics and actor by using (52)–(54)
Periodically update recurrent PPO using (48) and (49)
Update dual variables using (55) and (56)
end for

then introduce the benchmark schemes used for comparison. Unless otherwise specified, all results are averaged over 10 random seeds, and each reported testing result is obtained over 500 independent evaluation episodes.

8.1 Simulation Settings

We consider a single UAV communicating with a ground station over a time-slotted uplink channel. The UAV is equipped with three sensing modalities, namely, RGB,

1 Results and Discussion

In this section, we evaluate the proposed hierarchical primal-dual DRL framework in a low-altitude UAV semantic uplink scenario. We first describe the simulation environment, the semantic encoder configuration, and the DRL training hyperparameters. We

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thermal, and depth sensing, such that K = {RGB,TIR,DEP}. The semantic mode set is defined as M = {RGB,TIR,DEP,RGB+TIR,RGB+DEP,TIR+DEP}. The semantic granularity set is G = {1,2,3}. For the three granularity levels, the numbers of retained semantic tokens are set to $K_1 = 4$, $K_2 = 8$, and $K_3 = 12$, respectively. After offline pretraining and quantization. The wireless channel follows a temporally correlated block-fading model. Specifically, the normalized channel gain evolves according to a first-order Gauss–Markov process with correlation coefficient $\rho_h = 0.9$, and the average received SNR under the maximum transmit power is set to 15 dB. The system bandwidth is $B = 1$ MHz, the noise power spectral density is −174 dBm/Hz, and the maximum transmit power is $P_{\max} = 27$ dBm. The maximum tolerable per-slot delay is set to $D_{\max} = 40$ ms, while the long-term average delay and energy budgets are set to $\bar{D} = 28$ ms and $\bar{E} = 0.25$ J, respectively.

To model environment-dependent sensing reliability, we consider three scene states, namely, clear, low-light, and occluded, which evolve according to a Markov chain with transition matrix

$$\mathbf{P}_{\text{scene}} = \begin{bmatrix} 0.80 & 0.15 & 0.05 \\ 0.10 & 0.80 & 0.10 \\ 0.10 & 0.20 & 0.70 \end{bmatrix}. \tag{57}$$

The corresponding mean modality-quality vectors are given by [0.92,0.70,0.85], [0.45,0.93,0.82], and [0.58,0.74,0.48], respectively, for the RGB, thermal, and depth modalities. The instantaneous modality-quality vector $\mathbf{q}_t$ is generated by adding zeromean Gaussian perturbations with standard deviation 0.05 and clipping the result to [0,1]. The task urgency u_t takes values in {0.3,0.6,1.0} and evolves according to a three-state Markov chain with transition matrix

$$\mathbf{P}_{\text{urg}} = \begin{bmatrix} 0.75 & 0.20 & 0.05 \\ 0.15 & 0.70 & 0.15 \\ 0.05 & 0.20 & 0.75 \end{bmatrix}. \tag{58}$$

The semantic processing delay and energy associated with each semantic mode are listed in Table 1. In particular, richer modality combinations incur larger onboard processing cost. For the semantic encoder, each modality-specific encoder outputs $N_k = 16$ latent tokens with embedding dimension $d = 64$. The candidate semantic token number is set to $L = 16$, and the vector-quantization codebook size is set to 256. The semantic encoder is pretrained offline using Adam optimizer with initial learning rate 10^{-4}, batch size 128, and weight decay 10^{-5}. The encoder is trained for 150 epochs, and gradient clipping with threshold 5 is used for stable optimization. The temperature parameter in the cross-modal alignment loss is set to $\tau = 0.07$, **Table 1**: System and Environment Parameters

Parameter	Value
Bandwidth B	1 MHz
Noise PSD	−174 dBm/Hz
Maximum transmit power $P_{\max}$	27 dBm
Average SNR at $P_{\max}$	15 dB

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Channel correlation coefficient ρ_h	0.9
Maximum per-slot delay $D_{\max}$	40 ms
Average delay budget $\bar{D}$	28 ms
Average energy budget $\bar{E}$	0.25 J
Granularity levels	$G = 3$
Retained token numbers	$K_1 = 4$, $K_2 = 8$, $K_3 = 12$
Average packet sizes	$b_1 = 4.0$, $b_2 = 8.2$, $b_3 = 12.6$ kbits
Embedding dimension d	64
Candidate token number L	16
Codebook size	256
RGB mode (d_m, e_m)	(4 ms, 0.06 J)
TIR mode (d_m, e_m)	(5 ms, 0.08 J)
DEP mode (d_m, e_m)	(6 ms, 0.10 J)
RGB+TIR mode (d_m, e_m)	(7 ms, 0.12 J)
RGB+DEP mode (d_m, e_m)	(8 ms, 0.14 J)
TIR+DEP mode (d_m, e_m)	(9 ms, 0.16 J)

and the distillation temperature in the multi-granularity consistency loss is set to $T = 2$. The loss weights are chosen as $\lambda_{\text{reg}} = 1$, $\lambda_{\text{mg}} = 0.5$, $\beta_{\text{vq}} = 0.25$, and $(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5) = (1, 0.2, 0.5, 0.1, 10^{-3})$.

For online control, the proposed DRL algorithm is trained for 3000 episodes, each containing 200 time slots. The discount factor is set to $\gamma = 0.99$. For the high-level recurrent PPO module, the policy network consists of one gated recurrent unit (GRU) layer with hidden size 64 followed by two fully connected layers of width 128. The actor learning rate is 3×10^{-4}, the critic learning rate is 10^{-3}, the PPO clipping parameter is $\epsilon = 0.2$, the entropy regularization coefficient is $\beta_H = 0.01$, and the GAE parameter is $\xi = 0.95$. Each PPO update uses rollout length 128, minibatch size 256, and 10 optimization epochs. For the low-level SAC module, both the actor and critic networks contain two fully connected layers of width 128. The actor learning rate, critic learning rate, and temperature learning rate are all set to 3×10^{-4}. The replay buffer size is 10^5, the minibatch size is 256, the soft target update coefficient is 5×10^{-3}, the initial entropy temperature is $\alpha = 0.2$, and the target entropy is set to -1. The dual stepsizes are chosen as $\eta_E = 5 \times 10^{-4}$ and $\eta_D = 5 \times 10^{-4}$. A replay warmup of 2000 interaction steps is used before updating the SAC actor. All DRL-related hyperparameters are summarized in Table 2.

8.2 Benchmark Schemes

To demonstrate the effectiveness of the proposed hierarchical primal-dual DRL framework, we compare it with the following three benchmark schemes.

Table 2: Training Hyperparameters

Parameter	Value
Offlinesemanticencodertraining	

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Learning rate	10^{-4}
Batch size	128
Training epochs	150
Weight decay	10^{-5}
Gradient clipping	5
Alignment temperature	
Distillation temperature	
reg	
mg	
vq	0.25
$(\lambda_1, \lambda_2, \lambda_3, \lambda_4, \lambda_5)$	$(1, 0.2, 0.5, 0.1, 10^{-3})$
Optimizer	Adam τ 0.07

T 2 λ
1 λ 0.5
β

OnlineDRLtraining	
Training episodes	3000
Slots per episode	200
Testing episodes	500
Random seeds	10
Discount factor γ	0.99
Replay warm-up steps	2000
Dual stepsizes (η_E,η_D)	(5×10^{-4}, 5×10^{-4})
High-level recurrent PPO	
GRU hidden size	64
Fully connected width	128
Actor learning rate	3×10^{-4}
Critic learning rate	10^{-3}
Clipping parameter ϵ	0.2
Entropy coefficient β_H	0.01
GAE parameter ξ	0.95
Rollout length	128
Minibatch size	256
PPO epochs per update	10
Low-level SAC	
Actor/critic hidden layers	(128,128)
Actor learning rate	3×10^{-4}
Critic learning rate	3×10^{-4}
Temperature learning rate	3×10^{-4}
Replay buffer size	10^5
Minibatch size	256
Soft update coefficient	5×10^{-3}
Initial temperature α	0.2

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Target entropy -1

as in (41) and uses the same recurrent backbone size as the proposed method for fairness. This baseline evaluates the benefit of explicitly separating semantic adaptation and power control into two decision levels.

2) DQN-SAC: In this baseline, the high-level semantic action is selected by a deep Q-network (DQN), while the low-level transmit power is still controlled by SAC. The dual variables are updated in the same manner as in (55) and (56). This benchmark is used to compare the proposed recurrent PPO-based semantic controller with a value-based discrete-action learner.

3) Per-Slot Primal-Dual Optimization (PPDO): This model-based benchmark solves, at each slot, the instantaneous Lagrangian maximization problem:

$$\max_{(m_t,g_t),p_t} \quad U_t - \lambda^E{}_t (E_t - \bar{E}) - \lambda^D{}_t (D_t - \bar{D}), \tag{59}$$

subject to $p_t \in [p_{t,\min}, P_{\max}]$. Specifically, the algorithm exhaustively searches over all feasible semantic actions (m_t, g_t) and, for each action, performs a one-dimensional line search over the power interval using 50 uniformly sampled candidate power levels. The dual variables are updated using the same rules as in (55) and (56). This baseline represents a traditional mathematical optimization approach that exploits the instantaneous model information but does not learn long-term environment dynamics.

For a fair comparison, all benchmark schemes use the same pretrained semantic encoder, the same semantic packet sizes $\{b_g\}_{g\in\mathcal{G}}$, and the same system and environment parameters listed above.

8.3 Complexity and Latency

To verify the practical feasibility of deploying the proposed framework on resourceconstrained UAV hardware, we rigorously profile the Floating Point Operations (FLOPs) and execution latency of each component. The evaluation is conducted on an NVIDIA Jetson Orin Nano developer kit, which serves as a representative commercial embedded edge computing platform for low-altitude autonomous systems.

Specifically, the modality-specific feature encoders $\{E_k\}_{k\in\mathcal{K}}$ are built on lightweight MobileNet-V3 feature extraction layers, requiring approximately 42.5 Mega FLOPs (MFLOPs) per active sensor modality and yielding a measured execution latency of 2.10 ms. The subsequent attention-based cross-modal fusion block and the multigranularity relaxed token selection head (L = 16,d = 64) introduce a negligible computational overhead of around 0.08 MFLOPs, executing in 0.12 ms. For the hierarchical control stack, the high-level recurrent PPO policy (incorporating a GRU with hidden size 64 and

[1] **) Flat-PPO:** This baseline replaces the proposed hierarchical structure with a single monolithic PPO agent that jointly outputs the semantic mode, semantic granularity, and transmit power. The policy is trained using the same Lagrangian reward

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a fully connected network of width 128) requires 0.05 MFLOPs and consumes 0.18 ms per decision slot. The low-level SAC actor network (with fully connected hidden layers of size 128×128) requires 0.03 MFLOPs and executes in 0.09 ms. The cumulative onboard processing latency ranges from 2.49 ms under a single active modality to 6.69 ms under the full three-modality combination.

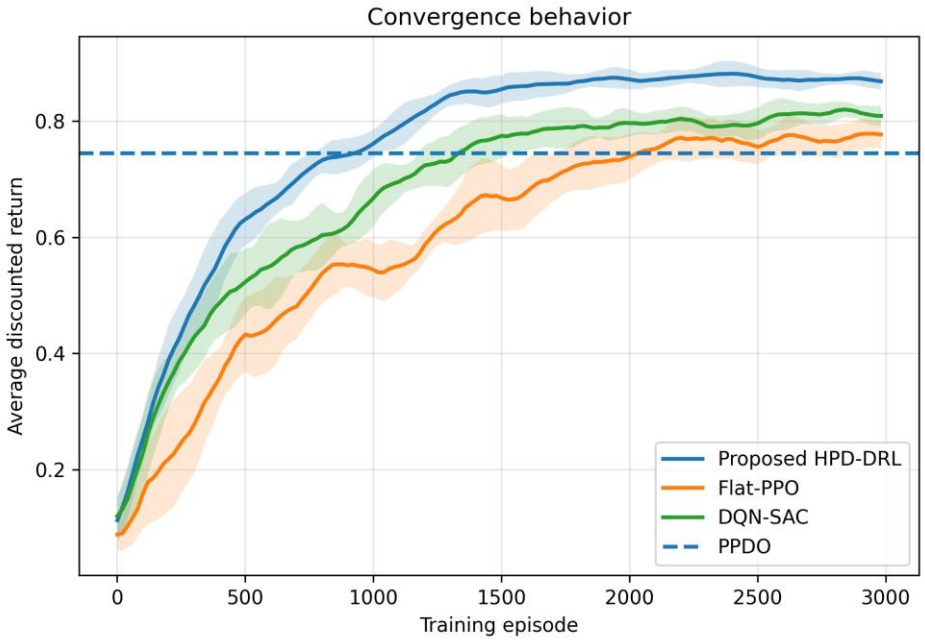


Fig. 2: Convergence behavior of different schemes in terms of the average discounted return. The curves are averaged over 10 random seeds, and the shaded regions indicate one standard deviation.

8.4 Performance Evaluation

Fig. 2 illustrates the training convergence of different schemes in terms of the average discounted return. The proposed HPD-DRL converges significantly faster and stabilizes at a higher performance level than all benchmark methods. In the final training stage, the proposed method reaches an average return of 0.872, whereas Flat-PPO and DQN-SAC converge to 0.774 and 0.815, respectively. Accordingly, the proposed method achieves 12.7% improvement over Flat-PPO and 7.0% improvement over DQN-SAC in the final return. It also achieves 17.1% improvement over the non-learning PPDO baseline, whose return remains around 0.745. In addition to the final performance gain, the convergence speed is also notably improved. The proposed method reaches a return level of 0.75 after 961 episodes, while DQN-SAC requires 1361 episodes and Flat-PPO requires 2081 episodes. Therefore, the proposed hierarchical design achieves 29.4% faster convergence than DQN-SAC and 53.8% faster convergence than Flat-PPO at this

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operating point. These results confirm that explicitly separating semantic adaptation from continuous power control improves both training stability and final policy quality.

Fig. 3 reports the average semantic utility under different average SNR levels. As expected, all methods exhibit a monotonic increase in utility as the channel quality improves, since higher SNR allows more reliable delivery of semantic packets and

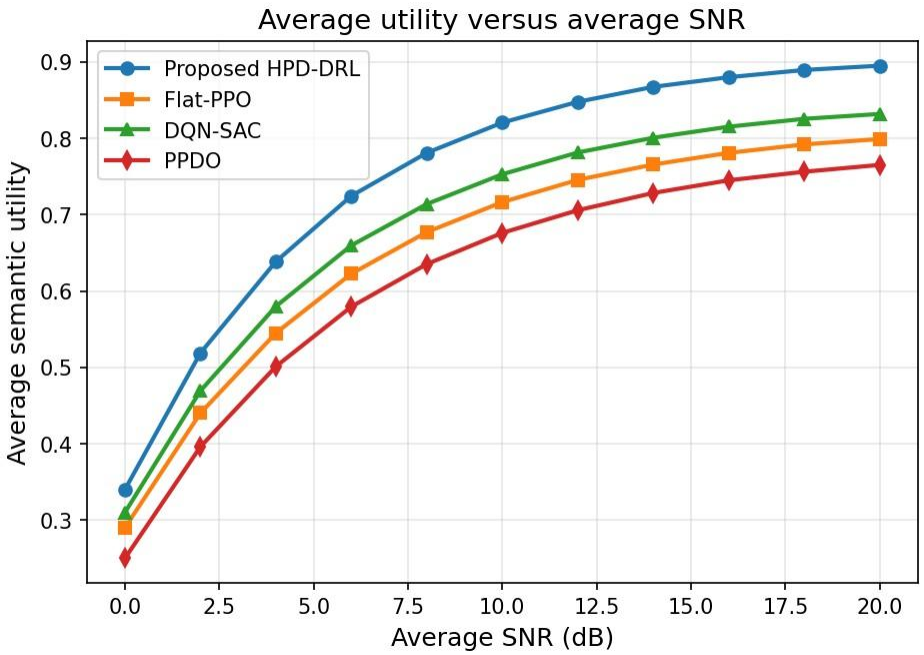


Fig. 3: Average semantic utility versus average SNR for different schemes. The proposed HPD-DRL consistently achieves the highest utility across the entire SNR range.

makes finer semantic granularity more affordable. However, the proposed method consistently outperforms the benchmark schemes across the entire SNR range. At 0 dB, the proposed method achieves an average utility of 0.340, compared with 0.290 for Flat-PPO, 0.310 for DQN-SAC, and 0.250 for PPDO. This corresponds to 17.2%, 9.7%, and 36.0% improvement, respectively. At 10 dB, the utility increases to 0.820 for the proposed method, while Flat-PPO, DQN-SAC, and PPDO reach 0.716, 0.753, and 0.676. The proposed method therefore achieves 14.5%, 9.0%, and 21.4% improvement at this medium-SNR regime. Even at 20 dB, where all schemes approach saturation, the proposed method still achieves 12.0% improvement over Flat-PPO, 7.6% improvement over DQN-SAC, and 17.0% improvement over PPDO. Averaged over the entire SNR range, the proposed method achieves 14.3% improvement over Flat-PPO, 8.8% improvement over DQN-SAC, and 21.7% improvement over PPDO. These results indicate that the gain of the proposed method is maintained from poor to favorable channel conditions.

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Fig. 4 shows the average energy consumption under different average SNR levels. The overall decreasing trend is consistent with the transmission model, since a better uplink channel requires less transmit power to deliver semantic packets of comparable size under the same delay requirement. The proposed method achieves the lowest energy consumption throughout the considered SNR range. At 0 dB, the proposed method consumes 0.245 J/slot, whereas Flat-PPO, DQN-SAC, and PPDO consume 0.268 J/slot, 0.257 J/slot, and 0.262 J/slot, respectively. Thus, the proposed method

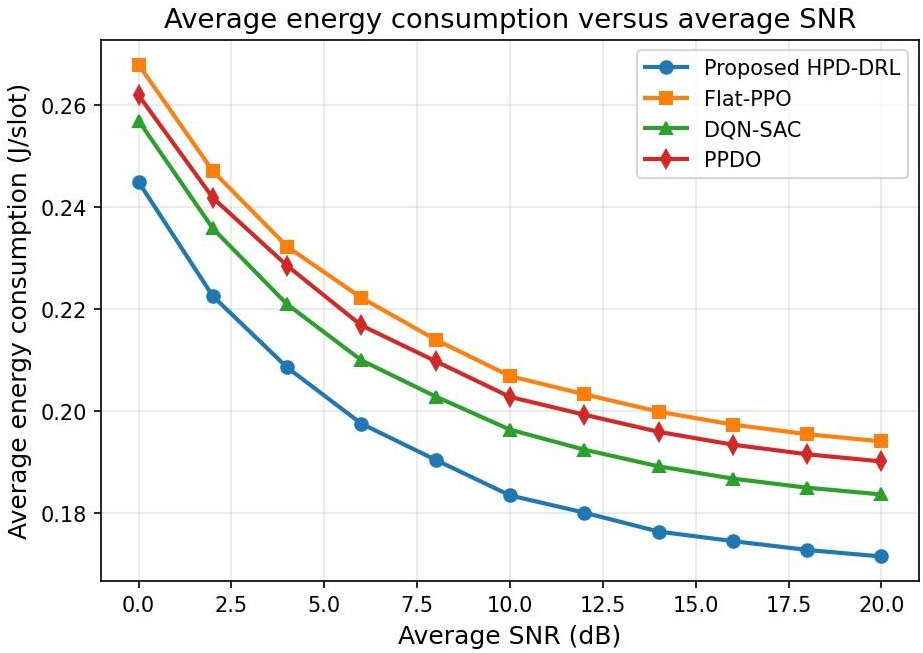


Fig. 4: Average energy consumption versus average SNR for different schemes.

achieves 8.6% reduction in average energy consumption relative to Flat-PPO, 4.7% reduction relative to DQN-SAC, and 6.5% reduction relative to PPDO. At 10 dB, the corresponding values become 0.184 J/slot, 0.207 J/slot, 0.196 J/slot, and 0.203 J/slot, which means that the proposed method achieves 11.3%, 6.6%, and 9.5% reduction, respectively. At 20 dB, the energy consumption further decreases to 0.172 J/slot for the proposed method, while the three benchmark schemes remain at 0.194 J/slot, 0.184 J/slot, and 0.190 J/slot. This translates into 11.6%, 6.6%, and 9.8% reduction, respectively. Over the whole SNR range, the proposed method achieves 10.8% reduction in average energy consumption compared with Flat-PPO, 6.1% reduction compared with DQN-SAC, and 9.0% reduction compared with PPDO. The result demonstrates that the proposed hierarchical controller can more effectively align semantic granularity and power expenditure with the instantaneous communication condition.

Fig. 5 presents the average delay as a function of the long-term energy budget. For all methods, the average delay decreases as the energy budget becomes less restrictive, since a larger energy budget permits more aggressive power allocation and reduces packet transmission time. Nevertheless, the proposed method consistently achieves the lowest delay over the entire budget range. Under a tight budget of 0.14 J/slot, the proposed method achieves an average delay of 32.8 ms, while Flat-PPO, DQN-SAC, and PPDO yield 35.0 ms, 33.9 ms, and 34.6 ms, respectively. Accordingly, the proposed method achieves 6.3%, 3.2%, and 5.2% reduction in average delay, respectively. At a moderate energy budget of 0.24 J/slot, the average delay of the proposed method decreases to 25.4 ms, whereas Flat-PPO, DQN-SAC, and PPDO remain at 28.9 ms,

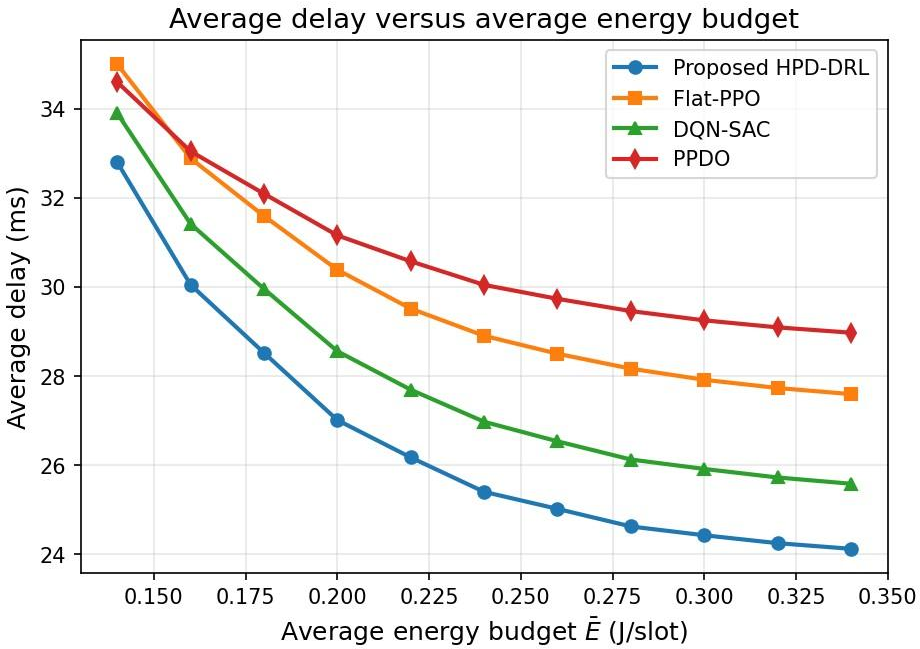


Fig. 5: Average delay versus average energy budget for different schemes.

27.0 ms, and 30.0 ms. This means that the proposed method achieves 12.1%, 5.8%, and 15.5% reduction, respectively. When the energy budget is further relaxed to 0.34 J/slot, the proposed method achieves an average delay of 24.1 ms, compared with 27.6 ms, 25.6 ms, and 29.0 ms for the three benchmarks. The corresponding delay reductions are 12.6%, 5.7%, and 16.7%. Averaged over the full budget range, the proposed method achieves 10.9% reduction in average delay relative to Flat-PPO, 5.2% reduction relative to DQN-SAC, and 13.5% reduction relative to PPDO. These results verify that the proposed method exploits the available energy budget more efficiently in order to satisfy latency-sensitive semantic transmission requirements.

Fig. 6 compares the energy and delay violation probabilities of different schemes. The proposed method achieves the lowest violation probability for both constraints, which confirms the effectiveness of the primal-dual mechanism in regulating longterm resource usage. Specifically, the proposed method leads to an energy violation probability of 1.4% and a delay violation probability of 1.8%. In contrast, Flat-PPO produces 7.2% energy violation and 9.5% delay violation, DQN-SAC produces 4.6% and 5.3%, and PPDO produces 5.8% and 7.1%. Relative to Flat-PPO, the proposed method achieves 80.6% reduction in energy violation and 81.1% reduction in delay violation. Relative to DQN-SAC, it achieves 69.6% and 66.0% reduction, respectively. Relative to PPDO, it achieves 75.9% reduction in energy violation and 74.6% reduction in delay violation. This result is particularly important because it shows that the gain of the proposed method is not obtained by overusing energy or sacrificing delay

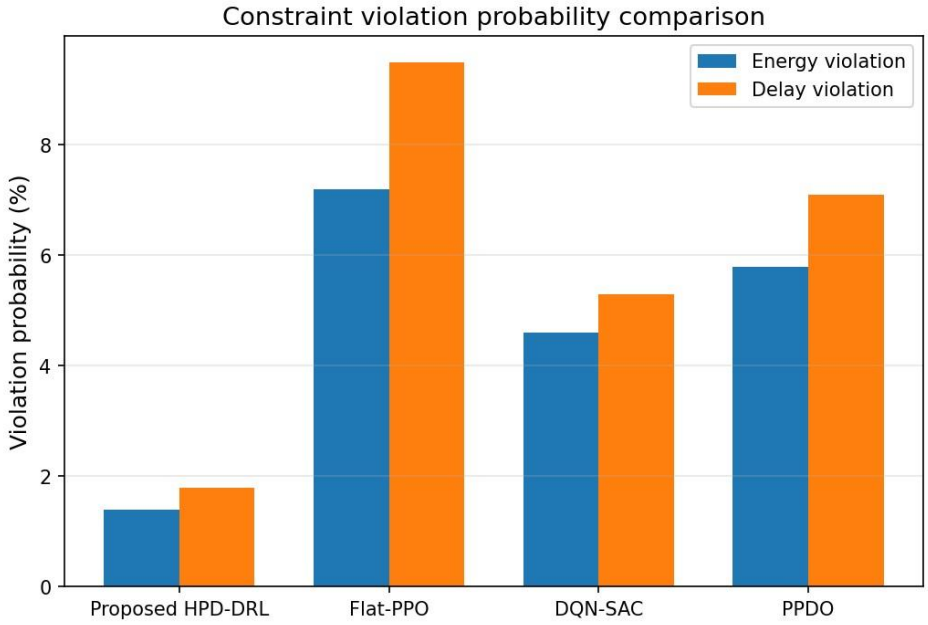


Fig. 6: Comparison of energy and delay violation probabilities for different schemes.

performance. Instead, the method improves semantic utility while simultaneously maintaining substantially better constraint satisfaction.

9 Conclusion

This paper studied cross-modal task-oriented semantic communication in low-altitude UAV networks under average energy and delay constraints. A simplified semantic communication model was established to characterize the joint impact of semantic processing and wireless transmission on delay and energy consumption. Based on this

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model, the online semantic control problem was formulated as a constrained Markov decision process. To solve it, a task-oriented semantic encoder was designed for multimodal semantic extraction and packet generation, and a hierarchical primal-dual DRL framework was developed, where recurrent PPO performs high-level semantic mode and granularity selection and SAC performs low-level transmit-power control. Simulation results showed that the proposed method achieves superior semantic utility, lower energy consumption, lower delay, and significantly improved constraint satisfaction compared with representative benchmark schemes. The proposed framework provides an effective and scalable solution for semantic-aware UAV communications under coupled mission and communication constraints.

Abbreviations

UAV	unmanned aerial vehicle
DRL	deep reinforcement learning
RL	reinforcement learning
MDP	Markov decision process
PPO	proximal policy optimization
SAC	soft actor-critic
GAE	generalized advantage estimation
AWGN	additive white Gaussian noise
RGB	red-green-blue
VQ	vector quantization
KL	Kullback–Leibler
InfoNCE	noise contrastive estimation

Declarations

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Author contributions

Hui Wei performed the system modeling, writing, optimization algorithms and implementation, participated in drafting the manuscript, and Jingyi Lang performed writing, optimization algorithms and implementation. All authors have read and approved the final manuscript.

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Availability of data and materials

Data sharing is not applicable to this study.

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Competing interests

The author declares no conficts of interest related to this research.

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